

Shaocong Ma | Curriculum Vitae

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Education & Training

- ❖ **Postdoctoral Associate** 2024–present
University of Maryland, College Park
Advisor: Prof. Heng Huang
- ❖ **Ph.D. in Electrical and Computer Engineering** 2019–2024
University of Utah
Advisor: Prof. Yi Zhou
- ❖ **M.A. in Statistics** 2017–2019
University of California, Santa Barbara
- ❖ **B.S. in Statistics** 2013–2017
Sichuan University

Academic & Research Experiences

- ❖ **Research Internship**, Lawrence Livermore National Laboratory May 2022–Aug. 2022
Advisor: Dr. Bhavya Kailkhura, Dr. James Diffenderfer
 - Implemented high-performance parallelized graph-based physics-informed neural networks (PINNs) & neural operators for computational fluid dynamics (CFDs).
- ❖ **Research Assistant**, University of Utah Aug. 2019–May 2024
Advisor: Prof. Yi Zhou
 - Developed robust and efficient reinforcement learning (RL) and stochastic optimization theory.

Preprint & Submission

- [1] **Shaocong Ma**, Weidong Cai, and Heng Huang. “Hyper-Octant Zeroth-Order Optimization: Fine-Tuning Quantized LLMs on the Positive Orthant Manifold”. Submitted to NeurIPS. 2026.
- [2] **Shaocong Ma**, Chenhui Xu, Jinjun Xiong, and Heng Huang. “A Multi-Fidelity Mixture-of-Expert Framework Integrating PDE Solvers and Neural Operators for Computational Fluid Dynamics”. Submitted to NeurIPS. 2026.
- [3] Ziyi Chen, **Shaocong Ma**, and Heng Huang. “Zeroth-Order Algorithm with the Best Direction”. Submitted to NeurIPS. 2026.
- [4] Qi He, **Shaocong Ma**, and Heng Huang. “From Noisy Zeroth-Order Updates to Stable Descent: Provable Stabilization by Shrink-or-Fallback”. Submitted to NeurIPS. 2026.
- [5] **Shaocong Ma**, Qi He, and Heng Huang. “Dual Clipped Zeroth-Order Optimization for Skewed Noise Oracles”. Submitted to NeurIPS. 2026.
- [6] **Shaocong Ma**, Su Zhang, and Heng Huang. “MI-ZOPG: Randomized Multi-Index Monte Carlo Estimation for Zeroth-Order Policy Gradient”. Submitted to NeurIPS. 2026.

Publications

- [7] **Shaocong Ma**, Peiran Yu, and Heng Huang. “New Hybrid Fine-Tuning Paradigm for LLMs: Algorithm Design and Convergence Analysis Framework”. In: *International Conference on Learning Representations (ICLR)*. 2026.
- [8] **Shaocong Ma** and Heng Huang. “Riemannian Zeroth-Order Gradient Estimation with Structure-Preserving Metrics for Geodesically Incomplete Manifolds”. In: *International Conference on Learning Representations (ICLR)*. 2026.

- [9] **Shaocong Ma** and Heng Huang. “On the Optimal Construction of Unbiased Gradient Estimators for Zeroth-Order Optimization”. In: *Advances in Neural Information Processing Systems (NeurIPS)* (2025). *NeurIPS 2025 Spotlight*.
- [10] **Shaocong Ma** and Heng Huang. “Robust Reinforcement Learning in Finance: Modeling Market Impact with Elliptic Uncertainty Sets”. In: *Advances in Neural Information Processing Systems (NeurIPS)* (2025).
- [11] **Shaocong Ma** and Heng Huang. “Revisiting Zeroth-Order Optimization: Minimum-Variance Two-Point Estimators and Directionally Aligned Perturbations”. In: *International Conference on Learning Representations (ICLR)* (2025). *ICLR 2025 Spotlight*.
- [12] **Shaocong Ma**, Ziyi Chen, Yi Zhou, and Heng Huang. “Rectified Robust Policy Optimization for Model-Uncertain Constrained Reinforcement Learning without Strong Duality”. In: *Transactions on Machine Learning Research (TMLR)* (2025).
- [13] **Shaocong Ma**, James Diffenderfer, Bhavya Kailkhura, et al. “Deep learning of PDE correction and mesh adaption without automatic differentiation”. In: *Machine Learning* (2025).
- [14] Yi Zhou and **Shaocong Ma**. “Stochastic Optimization Methods for Policy Evaluation in Reinforcement Learning”. In: *Foundations and Trends in Optimization* (2024).
- [15] **Shaocong Ma**, Ziyi Chen, Shaofeng Zou, and Yi Zhou. “Decentralized Robust V-learning for Solving Markov Games with Model Uncertainty”. In: *The Journal of Machine Learning Research (JMLR)* (2023).
- [16] Ziyi Chen, **Shaocong Ma**, and Yi Zhou. “Accelerated proximal alternating gradient-descent-ascent for nonconvex minimax machine learning”. In: *2022 IEEE International Symposium on Information Theory (ISIT)* (2022).
- [17] Ziyi Chen, **Shaocong Ma**, and Yi Zhou. “Finding correlated equilibrium of constrained Markov game: A primal-dual approach”. In: *Advances in Neural Information Processing Systems (NeurIPS)* (2022).
- [18] **Shaocong Ma**, Ziyi Chen, Yi Zhou, Kaiyi Ji, and Yingbin Liang. “Data sampling affects the complexity of online sgd over dependent data”. In: *Uncertainty in Artificial Intelligence (UAI)* (2022).
- [19] Ziyi Chen, **Shaocong Ma**, and Yi Zhou. “Sample efficient stochastic policy extragradient algorithm for zero-sum markov game”. In: *International Conference on Learning Representations (ICLR)* (2021).
- [20] **Shaocong Ma**, Ziyi Chen, Yi Zhou, and Shaofeng Zou. “Greedy-GQ with Variance Reduction: Finite-time Analysis and Improved Complexity”. In: *International Conference on Learning Representations (ICLR)* (2020).
- [21] **Shaocong Ma** and Yi Zhou. “Understanding the impact of model incoherence on convergence of incremental SGD with random reshuffle”. In: *International Conference on Machine Learning (ICML)* (2020).
- [22] **Shaocong Ma**, Yi Zhou, and Shaofeng Zou. “Variance-reduced off-policy TDC learning: Non-asymptotic convergence analysis”. In: *Advances in Neural Information Processing Systems (NeurIPS)* (2020).

Successful Grant Writing Experiences

- ❖ Advanced AI Framework to Improve Understanding and Prediction of Wildland Fire 2026–2028
Awarded by NSF-RISE, **\$1,856,577**.
- ❖ A Real-World Test Bed for Post-Market Surveillance and Stress Testing of AI-Enabled Imaging Tools 2025–2027
Awarded by FDA, **\$1.2M**.
- ❖ Ultrascale Machine Learning to Empower Discovery in Alzheimers Disease Biobanks 2026–2031
Recommended, NIH center grant, **\$15M**.
- ❖ Other grant writing experience for NSF MFAI, NSF GCR, NSF PCL, NSF SLES, NSF/NIH SCH, NIH R01s.

Honors & Awards

- ❖ ICML 2026, Sliver Reviewer Award 2026
- ❖ NeurIPS 2025, Top Reviewer Award 2025

❖ NeurIPS 2025, Spotlight (top 3% of submissions)	2025
❖ ICLR 2025, Spotlight (top 5% of submissions)	2025
❖ AISTATS 2025, Best Reviewer Award	2025
❖ Graduate 1st Place at University of Utah	2024
❖ Invited Monograph Contributor, Now Publishers (Foundations and Trends in Optimization)	2024
❖ Invited Participant (fully funded), NSF-Simons MoDL Annual Meeting, Simons Foundation, New York	2023
❖ Excellence Scholarship, The College of Mathematics, Sichuan University	2013–2017
❖ Second Prize, Chinese Mathematics Competitions for College Students, Sichuan Province Division	2014

Invited Talks

❖ New Hybrid Fine-Tuning Paradigm for LLMs: Algorithm Design and Convergence Analysis Framework Workshop on Mathematics of Dynamical Systems in Modern Machine Learning Brin Mathematics Research Center, University of Maryland	Sep. 2026
❖ Minimum-Variance Two-Point Estimators and Directionally Aligned Perturbations Modeling and Optimization: Theory and Applications (MOPTA) 2026 Department of Industrial and Systems Engineering, Lehigh University	Aug. 2026
❖ Towards Efficient Machine Learning: Algorithms, Theoretical Foundations, and Applications Department of Mathematics and Statistics, Utah State University Department of Computer Science, Texas Tech University Department of Computer Science, Kent State University	Apr. 2026 Feb. 2026 Feb. 2026

Teaching Experiences

❖ Coteach , University of Maryland, College Park ◦ CMSC422: Introduction to Machine Learning	2024–2026
❖ Teaching Assistant , University of Utah ◦ ECE 3500: Fundamentals of Signals and Systems	2020–2021
❖ Teaching Assistant , University of California, Santa Barbara ◦ PSTAT 5A: Statistics; ◦ PSTAT 5LS: Statistics for Life Science; ◦ PSTAT 109: Statistics for Economics; ◦ PSTAT 172A: Actuarial Statistics; ◦ PSTAT 175: Survival Analysis	2017–2019

Professional Services

❖ Conference Reviewer : ICML; ICLR; NeurIPS; IJCAI; UAI; AAAI; AISTATS; RLC; IEEE BigData; IEEE CDC.	
❖ Journal Reviewer : Transactions on Machine Learning Research (TMLR); IEEE Transactions on Signal Processing (TSP); IEEE Transactions on Networking (ToN); IEEE Transactions on Emerging Topics in Computational Intelligence (TETCI); Scientific Reports; Numerical Algorithms; Nonlinear Dynamics; European Journal of Control.	
❖ Workshop Reviewer : ICLR Blogpost.	